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Uncloneable encryption from decoupling

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The laws of quantum physics mean that prominent classical cryptographic protocols can be broken using quantum computers, but they also permit security guarantees that are classically impossible. For example, quantum states cannot be cloned, which restricts the capabilities of any adversary. Here we show that uncloneable encryption exists with no computational assumptions, with security approaching the ideal value as an inverse-polynomial function of the security parameter. With this scheme, two non-interacting adversaries cannot both learn an encrypted message, even if they are both given the encryption key. Our proof uses the properties of a monogamy-of-entanglement game associated with the Haar measure encryption. Using this connection, we show that any state that succeeds with high probability cannot be close to being maximally entangled between the referee and either of the adversaries. The decoupling principle then implies that either adversary becomes completely uncorrelated and, therefore, cannot win significantly better than random guessing.

Uncloneable cryptography is a paradigm wherein the no-cloning principle of quantum mechanics[1–3], which asserts that quantum data cannot be copied, is used to achieve classically impossible security guarantees. This underpins much of the groundbreaking work in quantum cryptography, notably quantum key distribution[4] and quantum money[5]. An uncloneable encryption[6] scheme encodes a classical message as a quantum ciphertext to guarantee that two non-interacting adversaries cannot both learn the message, even when given the encryption key. This defines a stronger, classically impossible notion of security, as any classical ciphertext can be copied. However, a security proof for uncloneable encryption has been elusive. Here we show the existence of an uncloneable bit in the information-theoretic model, with no computational assumptions. An uncloneable bit is a family of quantum encryptions of classical messages (QECMs) encoding a single bit that can be scaled to have arbitrarily good uncloneable security. The property that we refer to as uncloneable security, which we show for the uncloneable bit, is that, in the limit of large dimension, the success probability of a cloning attack must tend to 1/2. So far, only heuristic arguments for security have been shown[6–8], and various schemes[9–11] with unproven unconditional security have been proposed, to the extent that the possibility of uncloneable encryption has also been questioned[7,12], without conclusive answers. We prove the unconditional security of uncloneable encryption and demonstrate the existence of a stronger form of uncloneable cryptography, which is known to imply the security of quantum key distribution[13]. According to results in ref. 14, an uncloneable bit can be used to construct secure uncloneable encryption schemes for messages of arbitrary length. Hence, the uncloneable bit is a fundamental primitive in uncloneable cryptography and forms the basis for quantum copy protection[15–18], secure software leasing[19–21], quantum functional encryption[22], uncloneable decryption[17,23,24] and uncloneable zero-knowledge proofs[25].

An uncloneable bit consists of a QECM: a probability distribution μ over encryption keys k and, for each key, a pair of quantum states σ_0^k and σ_1^k representing the ciphertexts for the two possible single-bit messages. The encryption is correct if the message can be perfectly recovered from the encryption and the key, which is equivalent to the orthogonality of the message states for a given key. This property is implicitly required in this work. The security of an uncloneable bit is defined by means of a security game that models a cloning attack against the QECM and is played between an honest referee, Alice, and two cooperating malicious adversaries, Bob and Charlie. The adversaries attempt to clone the ciphertext state before being given the key by simultaneously but independently trying to decrypt the message.

(1) Alice samples a random key k from μ and a bit message $x \in \{0, 1\}$ uniformly and prepares the quantum ciphertext σ_x^k.

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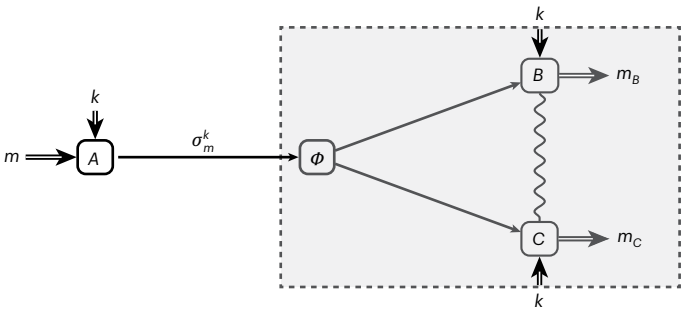


Fig. 1 | Cloning attack against a QECM. The parts relating to the cloning attack are outlined by a dotted box as defined in the text.

(2) σ_x^k passes through an adversarially chosen quantum channel, called the pirate channel, which outputs an entangled state in Bob's and Charlie's systems.
(3) Alice informs Bob and Charlie of the key k.
(4) Without communicating, Bob and Charlie both try to guess the original message x.
(5) The adversaries win if both of their guesses are correct.

This scenario is illustrated in Fig. 1. By making a random but coordinated guess of the message bit, Bob and Charlie can always win the game with probability 1/2. The level of security, called the uncloneable security, is quantified by how much better they can do than this in terms of the winning probability. For example, if Alice encodes the message in one of the two conjugate-coding bases, Bob and Charlie can win with probability $\cos^2(\pi/8) \approx 0.85$ by having the pirate channel measure in a basis interpolating between the two[13]. See Section 1.5 in ref. 26 for the formal definition of this model.

In this work, we study the Haar measure encryption of a bit, where, to encode a bit, Alice samples a random basis from the Haar measure on the unitary group of a fixed dimension d and prepares a randomly chosen state from among either the first or second half of the states in the basis, depending on the value of the bit. QECMs based on the Haar measure were first introduced in ref. 12 and have been further studied in ref. 27. We show that the uncloneable security of the Haar measure encryption scales as $O(\log(\log d)/\log d)$, which is inverse-polynomial in the number of qubits. We also make use of 2-designs to provide a distribution on the unitary group that can be efficiently sampled and achieves the same security guarantee with regard to the dimension. In terms of the security parameter λ, this translates to uncloneable security $\tilde{O}(1/\lambda)$. This differs from the definition first proposed in ref. 6, which requires the scaling to be negligible in λ—smaller than any inverse polynomial—but coincides with the notion of 'weak uncloneable security' introduced in ref. 11. The possibility of a negligible bound, which we refer to as strong uncloneable security, remains an open problem.

An important property used to study uncloneable encryption is its relationship to monogamy-of-entanglement (MoE) games. Monogamy is a property of quantum entanglement[28] that asserts that in a tripartite system consisting of systems held by Alice, Bob and Charlie, if Bob is highly entangled with Alice, then Charlie can only be weakly entangled. One of the ways in which the strength of this property is quantified is via the winning probability of an MoE game. An MoE game, first defined in ref. 13, is a game played between a referee, Alice, and two cooperating players, Bob and Charlie, as follows:

(1) Bob and Charlie prepare a state shared among themselves and Alice, and they are then separated and can no longer communicate.
(2) Alice samples a question θ and informs Bob and Charlie.
(3) Alice makes a measurement specified by θ to get answer x.
(4) Bob and Charlie measure their parts of the state and both attempt to guess x.
(5) The players win if both their guesses are correct.

See Section 1.5 in ref. 26 for the formal definition. Many forms of QECM, such as the Haar measure encryption, have an associated MoE game where cloning attacks can be mapped to strategies for the game in such a way that the success probability is preserved, as induced by the Choi–Jamiołkowski isomorphism[29]. This can be seen as a type of equivalence between prepare-and-measure and entanglement-based schemes. The Haar measure game, the MoE game analogue of Haar measure encryption, is illustrated in Fig. 2. Many known bounds on the cloning values of QECMs arise from studying the associated MoE game, for example using the overlap technique[13,30,31] or the Navascués–Pironio–Acín hierarchy[11,32]. However, none of the known techniques have been able to give tight enough upper bounds on the MoE game value to demonstrate uncloneable encryption.

We achieve this by means of decoupling, originally proposed in the context of error suppression[33], which is now a well-known paradigm in quantum information theory[34] with notable applications in quantum Shannon theory[35], resource theories[36] and, more recently, authentication[37] and quantum encryption[38] in cryptography. We make use of the one-shot refinement described in ref. 39. In a tripartite pure state ϕ_{ABC}, we say that systems A and C decouple if the reduced state of $\phi_{AC} = \phi_A \otimes \phi_C$ is a product state, in which case A is purified by subsystem B. This means that C does not provide any information on A, that is, the outcome of any measurement of A is statistically independent of the outcome of any measurement on C.

The relevance of decoupling in showing the validity of uncloneable encryption lies in proving that the quantum value of the Haar measure game is sufficiently bounded. Assume that Bob and Charlie can do substantially better at the Haar measure game than a coordinated random guess, which means that there exists a state ψ_{ABC} that fares well in this game such that its winning probability is well above 1/2. Then, the overlap between ψ_{ABC} and any state σ_{ABC} where systems A and C decouple is very small. Note that, by symmetry between Bob and Charlie, we could alternatively begin with a state where A and B decouple. By MoE, this would mean that the overlap of ψ_{ABC} and σ_{ABC} is also very small when systems A and B are maximally entangled. The decoupling inequality on ψ_{ABC} in this context says that the amount of independent randomness (not shared) that can be extracted from A depends on how far systems A and B are from being maximally entangled, quantified by the overlap with a maximally entangled state. As we know that this overlap is small for ψ_{ABC}, Alice's randomized measurement must result in her system being decoupled from B. In the context of decoupling, this measurement is equivalent to Alice applying a Haar-random unitary to her system and then tracing out a subsystem A_2 corresponding to all but the first qubit. She then measures the first qubit A_1. By virtue of A_1 decoupling from B, the probability of Bob correctly guessing Alice's measurement outcome is always low. However, our assumption implies, in contrast, that Bob must win, with probability much higher than 1/2. This is a contradiction. A schematic of this argument is given in Fig. 2. Hence, there cannot exist a state like ψ_{ABC}, which is the essence of Theorems 1 and 2, our main result.

Main result: unconditional security

We show that there exists a QECM that is correct and uncloneable secure, which is our main result, Theorem 1. The d-dimensional Haar measure encryption of a bit is denoted $Q_{d,2}$, and formally introduced in Definition 4 in ref. 26.

Theorem 1

The family of QECMs $\{Q_{2\lambda,2}\}_{\lambda\in\mathbb{N}}$ *is correct and uncloneable secure.*

See Definitions 3 and 6 in ref. 26 for the full definitions of correctness and uncloneable security. We show Theorem 1 by proving that the quantum value (or the winning probability) of the d-dimensional two-outcome Haar measure game $G_{d,2}$ (Definition 9 in ref. 26) is $1/2 + O(\log(\log d)/\log d)$, which is Theorem 2.

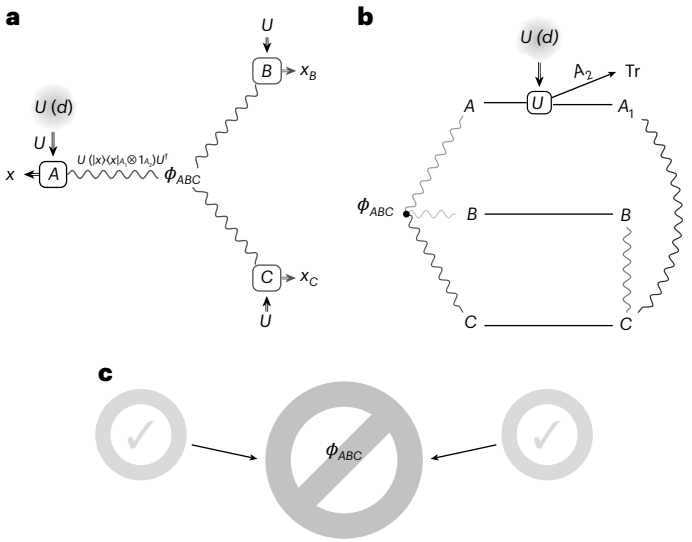


Fig. 2 | Decoupling implies the existence of uncloneable encryption. a, The Haar measure MoE game as described in the text. **b**, The decoupling theorem. **c**, Uncloneable encryption exists. It may seem like specific structure on the unitary U and the register A_2 is needed for decoupling to be applicable. However, this is not so, as the only information required is that U is Haar random, which is true by construction, and that the size of A_2 is sufficiently large compared with A_1, which naturally arises in decoupling.

Discussion

In this work, we have shown that uncloneable cryptography is possible without any computational assumptions. We studied the Haar measure encryption of a bit, which is the QECM where one bit of classical information is encoded as one of the two halves of a uniformly random basis. We showed that this has uncloneable, and hence uncloneable-indistinguishable, security with parameter $O(\log(\log d)/\log d)$. This constitutes a considerable advance over previous work, as uncloneable security with parameter tending to 0 was known only for the quantum random oracle model[6,8], and unconditional uncloneable security was known only with constant parameter ~0.098 (ref. 11). Further, we show that this security can be attained with an efficient construction by means of t-designs.

The central notion that enables uncloneable encryption is an inequality that provides a necessary and sufficient condition for which the unitary evolution of a system results in decoupling. Our main innovation is in an original use of this wherein it is guaranteed that a small enough and randomly chosen subsystem of a quantum system becomes uncorrelated with an adversarial environment. This guarantee holds as follows. Consider the prepare-and-measure picture where Alice sends mixed states corresponding to messages encoded in a QECM system. There may be a reference system R, up to an isometry, with which her state is highly entangled resulting in a purification ζ_{RA}. To achieve uncloneable encryption, Alice sends the purification of R to Bob through a noisy channel whose Stinespring dilation is the cloning isometry $V_{A\to BC}$ (without loss of generality, the scheme is symmetric under interchanging Bob and Charlie, so the cloning isometry can also be seen as the dilation of a noisy channel to Charlie). From our application of the decoupling theorem, in the resulting tripartite state ζ_{RBC}, the marginal state of RC decouples as $\rho_{RC} = \rho_R \otimes \rho_C$. Thus, B decomposes into subsystems $B = B_1B_2$ such that $\zeta_{RBC} = Z_B(\varphi_{RB_1} \otimes \vartheta_{B_2C})$, where Z_B is some unitary change of basis in B. This holds most generally, as all purifications are isometrically equivalent. Now, to decrypt, Bob constructs an isometric decoder $W_{B_1\to\tilde{B}}Z_B^\dagger$, which extracts the purification of R into Bob's preferred subsystem $\tilde{B}$. Again, by the isometric equivalence of purifications, Bob can choose his decoder to output $\zeta_{R\tilde{B}}$, as a result of which the input state of RA is successfully transmitted to $R\tilde{B}$ as desired. Therefore, Bob receives full information about A, so only he can recover the encoded message. In hindsight, the fact that the states used to encode messages must be highly mixed in uncloneable encryption is indicative of its inherent connection to a decoupling inequality.

Future directions

We show unconditional uncloneable security with a parameter that scales inverse-logarithmically in the dimension, and hence inverse-polynomially in the security parameter. To achieve the full strength of uncloneable cryptography, this should be improved to a negligible scaling in the security parameter. We believe that this tighter upper bound can be attained and leave it as an open question for future work.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41567-025-03154-7.

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Methods

In this section we state and provide an intuitive description of the technical results applied to achieve the main result.

Theorem 2

Let $d \geq 138$ be even. The quantum value of the d-dimensional two-outcome Haar measure game $G_{d,2}$ is

$$\mathfrak{w}(G_{d,2}) \leq \frac{1}{2} + \frac{\log(\log d)}{\log d}. \tag{1}$$

Note that this bound begins to be attained with an 8-qubit encryption, as $138 \leq 2^8$.

The proof of Theorem 1 then holds as a consequence of applying the one-shot decoupling theorem[39] stated in Theorem 3 of ref. 26.

In the proof of Theorem 1, the one-shot decoupling theorem is applied to show that one cannot win significantly better than random guessing in the above type of MoE game, where the $O(\log(\log d)/\log d)$ terms comes from the error term in the decoupling theorem. To see this, first note that the success probability of a cloning attack (Definition 6 in ref. 26) is upper bounded by the winning probability of a two-outcome d-dimensional Haar measure game, as stated in Lemma 3.

Lemma 3

Let $d \in \mathbb{N}$ be even. Then, the cloning value of $Q_{d,2}$ is upper bounded by the quantum value of $G_{d,2}$.

This result generalizes to a very wide class of QECM schemes. See, for example, Proposition 5.14 in ref. 29.

Note that in this section, we fix d and assume that we are working with a strategy for $G_{d,2}$, which we denote $S = (B, C, \{B_x^U\}, \{C_x^U\}, \rho_{ABC})$, where $\{B_x^U\}$ and $\{C_x^U\}$ are Bob's and Charlie's measurements on quantum systems B and C, respectively, and ρ_{ABC} is the shared quantum state. Next, we establish that the winning probability of this game is $1/2 + O(\log(\log d)/\log d)$ (Theorem 2) by contradiction. This is where the decoupling theorem plays a role. We bound the value of the conditional min-entropy away from its minimum value in Corollary 5 of Theorem 4 stated next, which appears as an exponential bound in the decoupling inequality (Theorem 3 in ref. 26).

Theorem 4

Let $P = \int \sum_{x\in\{0,1\}} U\Pi_x U^\dagger \otimes B_x^U \otimes C_x^U \, \mathrm{d}U$. Define $\varepsilon \in \mathbb{R}$ such that $\| P \| = 1/2 + \varepsilon$. Suppose $\rho_{ABC} = |\psi\rangle\langle\psi|_{ABC}$ is the maximum-eigenvalue eigenstate of P with eigenvalue $1/2 + \varepsilon$. Let $\sigma_{ABC} = |\phi\rangle\langle\phi|_{ABC}$ be such that $\sigma_{AC} = \omega_A \otimes \sigma_C$. Then, assuming $\varepsilon > 0$,

$$|\langle\psi|\phi\rangle|^2 \leq \frac{8}{2^{\varepsilon \log d}}. \tag{2}$$

Corollary 5

Let ρ be as in Theorem 4. Then,

$$H_{\min}(A|B)_\rho \geq -(1-\varepsilon)\log d - 3. \tag{3}$$

Furthermore, we contradict our initial assumption that there exists a state shared by Alice, Bob and Charlie that corresponds to a winning probability much greater than 1/2 in the game by showing that its overlap with any other tripartite state where Alice and Charlie decouple must be very small (Theorem 4). Then, by MoE, Alice and Bob are not highly entangled. Yet, this overlap being small implies that Alice's randomized measurements cause her system to become decoupled from Bob, such that the probability of Bob correctly guessing Alice's measurement outcome is always low, which negates our assumption.

We also show how to achieve a correct and uncloneable-secure QECM with security $\tilde{O}(1/\lambda)$ in the security parameter λ efficiently by a construction with unitary t-designs in Theorem 6.

For $n \in \mathbb{N}$, let $\mathcal{V}_n \subseteq \mathcal{U}(2^n)$ be a 2-design that can be efficiently implemented, which exists according to results in refs. 40,41. We denote the QECM where the unitaries are sampled from $\mathcal{V}_n$ rather than the full Haar distribution by $Q_{\mathcal{V}_n,2}$; this is formally defined in Definition 5 of ref. 26. Then, $\{Q_{\mathcal{V}_\lambda,2}\}_\lambda$ is an efficient QECM, and it is used in the following result.

Theorem 6

There exists an efficient QECM encoding a bit that is correct and $\tilde{O}(1/\lambda)$-uncloneable secure.

Inclusion and ethics statement

Both authors are equal contributors and stakeholders in this work. This research does not pose any risk to any individual or groups. This research was conducted via local collaboration at the host institution of the authors. One of the two authors represents a minority in STEM.

Data availability

No datasets were generated during this study.

Code availability

No code was generated or used in the analysis and performance of this work.

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Author contributions

Both authors contributed equally to all aspects of this article and to the writing of the paper.

Competing interests

The authors declare no competing interests.

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